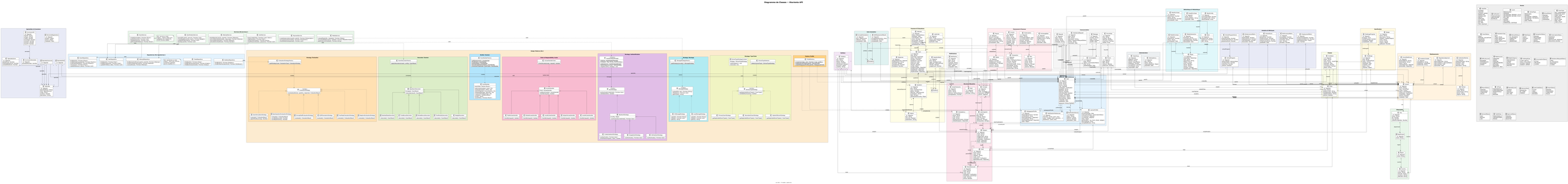

Diagramme de Classes — Xkorienta API

Spécialités & Orientation
CurriculumUE
_id: ObjectId
ueCode: String
ueType: UEType
credits: Number
hoursTotal: Number
hoursCM: Number
hoursTD: Number
hoursTP: Number
XkorientaRegistration
_id: ObjectId
student: Object
parent: Object
createdAt: Date
semester
SpecialtyScore
_id: ObjectId
globalScore: Number
employabilityScore: Number
accessibilityScore: Number
CurriculumSemester
_id: ObjectId
semesterNumber: Number
SpecialtyOutcome
_id: ObjectId
SpecialtySkill
_id: ObjectId
specialty
specialty
specialty
specialty
Specialty
_id: ObjectId
specialtyName: String
level: SpecialtyLevel
mode: String
duration: Number
description: String
Services (lib/services/)
ExamService
+createExam(data): Promise<IExam>
+publishExam(id): Promise<void>
+validateExam(id): Promise<void>
+getExamWithStats(id): Promise<IExam>
80+ services au total. Pattern : logique métier pure, délégation aux repositories, throw d'erreurs typées.
GamificationService
+awardXP(userId, amount): Promise<void>
+checkBadges(userId, event): Promise<Badge[]>
+updateStreak(userId): Promise<void>
+levelUp(userId): Promise<void>
AttemptService
+startAttempt(userId, examId): Promise<IAttempt>
+submitAttempt(attemptId, responses): Promise<IAttempt>
+resumeAttempt(token): Promise<IAttempt>
+calculateScore(attempt): Promise<Number>
AuthService
+login(credentials): Promise<{user, token}>
+register(data): Promise<IUser>
+refreshToken(token): Promise<String>
+resetPassword(token, password): Promise<void>
PaymentService
+createSubscription(userId, planId): Promise<ISubscription>
+processTransaction(data): Promise<ITransaction>
+handleWebhook(payload): Promise<void>
MediaService
+uploadMedia(file, metadata): Promise<IMedia>
+purchaseMedia(userId, mediaId): Promise<IMediaPurchase>
+getStreamUrl(mediaId): Promise<String>
uses
Repositories (lib/repositories/)
ExamRepository
+findById(id): Promise<IExam>
+findBySubject(subjectId): Promise<IExam[]>
+findPublished(filters): Promise<IExam[]>
+save(exam): Promise<IExam>
+updateStatus(id, status): Promise<void>
UserRepository
+findById(id): Promise<IUser>
+findByEmail(email): Promise<IUser>
+updateProfile(id, data): Promise<void>
AttemptRepository
+findByUserAndExam(userId, examId): Promise<IAttempt>
+findActiveAttempt(userId): Promise<IAttempt>
+save(attempt): Promise<IAttempt>
50+ repositories au total. Pattern : lean() pour lectures, projections explicites, pas de requêtes N+1.
ClassRepository
+findById(id): Promise<IClass>
+findByTeacher(teacherId): Promise<IClass[]>
+findBySchool(schoolId): Promise<IClass[]>
SyllabusRepository
+findById(id): Promise<ISyllabus>
+findByTeacher(teacherId): Promise<ISyllabus[]>
Design Patterns (lib/)
Strategy: Évaluation
EvaluationStrategyFactory
+getStrategy(type: EvaluationType): EvaluationStrategy
creates
«interface» EvaluationStrategy
+evaluate(attempt, question, response): EvaluationResult
ExamSimulationStrategy
+evaluate(): EvaluationResult
-applyPenalty(wrong): Number
OpenQuestionEvaluationStrategy
+evaluate(): EvaluationResult
-scoreByKeywords(): Number
-scoreBySemantic(): Number
ConceptSelfEvaluationStrategy
+evaluate(): EvaluationResult
QCMEvaluationStrategy
+evaluate(): EvaluationResult
TrueFalseEvaluationStrategy
+evaluate(): EvaluationResult
AdaptiveEvaluationStrategy
+evaluate(): EvaluationResult
Decorator: Examen
ExamDecoratorFactory
+applyDecorators(result, config): ExamResult
composes
BaseExamDecorator
#wrapped: ExamResult
+decorate(result): ExamResult
DetailedStatsDecorator
+decorate(): ExamResult
TimeBonusDecorator
+decorate(): ExamResult
StreakBonusDecorator
+decorate(): ExamResult
TimePenaltyDecorator
+decorate(): ExamResult
BadgeDecorator
+decorate(): ExamResult
Builder: Examen
ExamBuilderFactory
+selfAssessment(): ExamBuilder
+finalExam(): ExamBuilder
+practiceTest(): ExamBuilder
+fromTemplate(templateId): ExamBuilder
+fromExamType(type: ExamType): ExamBuilder
creates
ExamBuilder
-examData: Partial<IExam>
+setContext(subject, level): this
+setTarget(classes, fields): this
+setTiming(start, end, duration): this
+setMetadata(tags, difficulty): this
+customizeConfig(config): this
+validate(): ValidationResult
+build(): IExam
+saveDraft(): Promise<IExam>
Chain of Responsibility: Accès
AccessHandlerChain
+checkAccess(scope, request): boolean
builds chain
AccessHandler
#next: AccessHandler
+setNext(handler): AccessHandler
+handle(request): boolean
FieldAccessHandler
+handle(request): boolean
GlobalAccessHandler
+handle(request): boolean
LocalAccessHandler
+handle(request): boolean
SubjectAccessHandler
+handle(request): boolean
LevelAccessHandler
+handle(request): boolean
Strategy: Authentification
«Singleton» AuthStrategyManager
-instance: AuthStrategyManager
-strategies: Map<String, IAuthStrategy>
+getInstance(): AuthStrategyManager
+getStrategy(provider): IAuthStrategy
+register(name, strategy): void
manages
«interface» IAuthStrategy
+authenticate(credentials): Promise<User>
+validate(token): boolean
BaseAuthStrategy
#config: Object
+authenticate(credentials): Promise<User>
CredentialsAuthStrategy
+authenticate(): Promise<User>
-verifyPassword(plain, hash): boolean
GoogleAuthStrategy
+authenticate(): Promise<User>
GitHubAuthStrategy
+authenticate(): Promise<User>
Strategy: Stockage
StorageStrategyFactory
+getStrategy(provider): IStorageStrategy
selects
«interface» IStorageStrategy
+upload(file, key): Promise<String>
+delete(key): Promise<void>
+getUrl(key): String
S3StorageStrategy
+upload(file, key): Promise<String>
+delete(key): Promise<void>
+getUrl(key): String
LocalStorageStrategy
+upload(file, key): Promise<String>
+delete(key): Promise<void>
+getUrl(key): String
Strategy: Type École
SchoolTypeStrategyContext
-strategy: ISchoolTypeStrategy
+setStrategy(s): void
+execute(): any
SchoolTypeDetector
+detect(schoolType): ISchoolTypeStrategy
uses
creates
«interface» ISchoolTypeStrategy
+getApplicableExamTypes(): ExamType[]
+validateExam(exam): boolean
PrimaryExamStrategy
+getApplicableExamTypes(): ExamType[]
SecondaryExamStrategy
+getApplicableExamTypes(): ExamType[]
HigherEdExamStrategy
+getApplicableExamTypes(): ExamType[]
Factory: Profils
ProfileFactory
+createUser(data, role): Promise<{user, profile}>
-createLearnerProfile(userId): LearnerProfile
-createPedagogicalProfile(userId): PedagogicalProfile
Syllabus
Syllabus
_id: ObjectId
title: String
status: String
version: Number
isPublished: Boolean
structure: Object
visibility: String
Auto-évaluation
ConceptEvaluation
_id: ObjectId
level: MasteryLevel
reflection: String
evaluatedAt: Date
SelfAssessmentResult
_id: ObjectId
conceptAssessments: Array
overallScore: Number
completedAt: Date
Syllabus
Examens & Évaluations
Attempt
_id: ObjectId
status: AttemptStatus
startedAt: Date
submittedAt: Date
score: Number
percentage: Number
timeSpent: Number
resumeToken: String
guestSessionId: String
LateCode
_id: ObjectId
code: String
status: LateCodeStatus
usageHistory: Array
expiresAt: Date
Response
_id: ObjectId
selectedOptionId: ObjectId
textResponse: String
isCorrect: Boolean
pointsEarned: Number
timeSpent: Number
Exam
_id: ObjectId
title: String
examType: ExamType
status: ExamStatus
duration: Number
totalPoints: Number
startTime: Date
endTime: Date
isPublic: Boolean
config: Object
linkedConcepts: ObjectId[]
targetLevels: ObjectId[]
targetFields: ObjectId[]
targetClasses: ObjectId[]
QuestionBank
_id: ObjectId
text: String
type: EvaluationType
difficulty: DifficultyLevel
isPublic: Boolean
usageCount: Number
Question
_id: ObjectId
text: String
type: EvaluationType
points: Number
difficulty: DifficultyLevel
explanation: String
tags: String[]
openQuestionConfig: Object
Option
_id: ObjectId
text: String
isCorrect: Boolean
explanation: String
Syllabus
exam
responses
examId
questions
options
selectedOptionId
questionId
creates
competencies
Notifications
Notification
_id: ObjectId
type: String
title: String
message: String
isRead: Boolean
data: Object
createdAt: Date
Paiements & Finances
Payout
_id: ObjectId
amount: Number
status: PayoutStatus
recipientPhone: String
paymentProvider: String
processedAt: Date
Invoice
_id: ObjectId
type: String
status: String
amount: Number
currency: String
issuedAt: Date
dueDate: Date
pdfUrl: String
Subscription
_id: ObjectId
status: String
currentPeriodStart: Date
currentPeriodEnd: Date
amount: Number
autoRenew: Boolean
cancelledAt: Date
ExchangeRate
_id: ObjectId
baseCurrency: String
targetCurrency: String
rate: Number
source: String
fetchedAt: Date
walletId
relatedExam
transactionId
planId
Wallet
_id: ObjectId
balance: Number
totalEarned: Number
totalWithdrawn: Number
currency: String
isActive: Boolean
Transaction
_id: ObjectId
type: String
status: TransactionStatus
amount: Number
currency: String
paymentProvider: String
providerRef: String
productId: ObjectId
productType: String
Plan
_id: ObjectId
code: String
name: String
prices: Array
features: String[]
limits: Object
isActive: Boolean
Communication
Request
_id: ObjectId
type: String
status: String
priority: String
message: String
AssistanceRequest
_id: ObjectId
type: String
title: String
description: String
status: AssistanceRequestStatus
priority: String
resolvedAt: Date
Message
_id: ObjectId
content: String
type: String
isRead: Boolean
sentAt: Date
ForumPost
_id: ObjectId
title: String
content: String
isPinned: Boolean
viewCount: Number
replies: Array
conversationId
forumId
senderId
Conversation
_id: ObjectId
status: ConversationStatus
title: String
lastMessageAt: Date
Forum
_id: ObjectId
type: String
title: String
isActive: Boolean
postCount: Number
Administration
ImportLog
_id: ObjectId
fileName: String
type: String
status: ImportStatus
totalRows: Number
successRows: Number
errorDetails: Array
importedAt: Date
Idempotency
_id: ObjectId
key: String
status: String
responsePayload: Object
expiresAt: Date
Bibliothèque & Médiathèque
BookPurchase
_id: ObjectId
finalAmount: Number
status: String
downloadCount: Number
purchasedAt: Date
GuestPurchase
_id: ObjectId
email: String
downloadToken: String
expiresAt: Date
BookConfig
_id: ObjectId
commissionRate: Number
storageProvider: String
discountRules: Array
bookId
MediaPurchase
_id: ObjectId
finalAmount: Number
status: String
purchasedAt: Date
MediaInteraction
_id: ObjectId
eventType: String
completionRate: Number
watchedSeconds: Number
interactedAt: Date
Book
_id: ObjectId
bookFormat: String
pageCount: Number
isbn: String
mediaId
extends
Media
_id: ObjectId
mediaType: MediaType
title: String
description: String
fileKey: String
price: Number
isFree: Boolean
status: String
duration: Number
targetLevels: ObjectId[]
Analytics & Métriques
SchoolProgramScore
_id: ObjectId
matchScoreAvg: Number
valueForMoneyScore: Number
popularityScore: Number
InfrastructureMetric
_id: ObjectId
labsAvailable: Boolean
labQualityScore: Number
internetQuality: Number
libraryScore: Number
SchoolScore
_id: ObjectId
globalScore: Number
academicScore: Number
employmentScore: Number
infrastructureScore: Number
updatedAt: Date
PerformanceMetric
_id: ObjectId
successRate: Number
rankingPosition: Number
examType: String
year: Number
semester: Number
EmploymentMetric
_id: ObjectId
employmentRate6m: Number
employmentRate12m: Number
topEmployersJson: Object
year: Number
schoolProgram
SchoolProgram
_id: ObjectId
deliveryMode: DeliveryMode
durationYears: Number
tuitionFee: Number
isActive: Boolean
Gamification
ChallengeProgress
_id: ObjectId
progress: Array
overallProgress: Number
completed: Boolean
completedAt: Date
Badge
_id: ObjectId
name: String
description: String
category: BadgeCategory
rarity: String
iconUrl: String
criteria: Object
xpReward: Number
badges
challengeId
Challenge
_id: ObjectId
title: String
type: ChallengeType
status: String
startDate: Date
endDate: Date
goals: Array
rewards: Object
Classes
Invitation
_id: ObjectId
token: String
type: String
email: String
expiresAt: Date
maxUses: Number
currentUses: Number
status: String
targetClass
classId
Class
_id: ObjectId
name: String
academicYear: String
isIndependent: Boolean
validationStatus: String
isActive: Boolean
Établissements
UnverifiedSchool
_id: ObjectId
declaredName: String
declaredCity: String
status: String
matchedSchool: ObjectId
declaredBy: ObjectId[]
AcademicTutelle
_id: ObjectId
tutelleName: String
tutelleType: TutelleType
tutelleCode: String
RegulatoryApproval
_id: ObjectId
approvalNumber: String
approvalStatus: ApprovalStatus
approvalDate: Date
expiryDate: Date
InstitutionPartner
_id: ObjectId
relationshipType: String
startDate: Date
endDate: Date
description: String
partner
Partner
_id: ObjectId
name: String
partnerType: String
sector: String
logoUrl: String
School
_id: ObjectId
name: String
type: SchoolType
cycles: Cycle[]
status: SchoolStatus
subSystem: SubSystem
contactInfo: Object
isActive: Boolean
createdAt: Date
city
country
Géographie
City
_id: ObjectId
name: String
lat: Number
lng: Number
costOfLivingIndex: Number
department
Department
_id: ObjectId
name: String
region
Region
_id: ObjectId
name: String
country
Country
_id: ObjectId
name: String
isoCode: String
currency: String
dialCode: String
Utilisateurs
User
_id: ObjectId
name: String
email: String
phone: String
role: UserRole
subSystem: SubSystem
googleId: String
githubId: String
isActive: Boolean
isEmailVerified: Boolean
schools: ObjectId[]
preferences: Object
gamification: Object
paymentInfo: Object
createdAt: Date
updatedAt: Date
password: String {hashed}
preferences
devices
learnerProfile
pedagogicalProfile
PedagogicalProfile
_id: ObjectId
teachingExperience: Number
certifications: Array
teachingSubjects: ObjectId[]
accessScope: String
teachingSyllabuses: ObjectId[]
stats: Object
NotificationPreferences
_id: ObjectId
channels: {email, push, sms}
types: {badge, exam, payment, ...}
quietHours: {start, end}
NotificationDevice
_id: ObjectId
token: String
platform: String
isActive: Boolean
lastUsedAt: Date
LearnerProfile
_id: ObjectId
enrollmentDate: Date
subscriptionStatus: SubscriptionStatus
currentLevel: ObjectId
currentField: ObjectId
stats: Object
gamification: {xp, level, streak, badges}
learningGoals: Array
completedExams: Array
mainTeacher
students
teachers
relatedClass
relatedSubject
targetSubject
currentField
currentLevel
teachingSubjects
field
level
subject
Structure Éducative
CareerOutcome
_id: ObjectId
name: String
sector: String
demand: DemandStatus
averageSalary: Number
Skill
_id: ObjectId
name: String
skillType: String
Concept
_id: ObjectId
title: String
order: Number
keywords: String[]
parent: ObjectId
parent
Competency
_id: ObjectId
name: String
code: String
type: String
assessmentCriteria: String[]
relatedSubjects: ObjectId[]
LearningUnit
_id: ObjectId
title: String
unitType: String
order: Number
objectives: String[]
prerequisites: String[]
learningUnit
parentUnit
relatedSubjects
Subject
_id: ObjectId
name: String
code: String
subjectType: String
coefficient: Number
subSystem: SubSystem
levels: ObjectId[]
fields: ObjectId[]
fields
levels
parentField
applicableLevels
Field
_id: ObjectId
name: String
code: String
category: String
cycle: Cycle
subSystem: SubSystem
applicableLevels: ObjectId[]
EducationLevel
_id: ObjectId
name: String
code: String
cycle: Cycle
subSystem: SubSystem
schoolType: SchoolType
order: Number
isFinal: Boolean
Enums
UserRole
STUDENT
TEACHER
SCHOOL_ADMIN
INSPECTOR
SURVEILLANT
PREFET
PRINCIPAL
DG_ISIMMA
RECTOR
DG_M4M
TECH_SUPPORT
SubSystem
FRANCOPHONE
ANGLOPHONE
BILINGUAL
Cycle
PRESCOLAIRE
PRIMAIRE
SECONDAIRE_PREMIER_CYCLE
SECONDAIRE_SECOND_CYCLE
BTS_HND
LICENCE
MASTER
DOCTORAT
FORMATION_CONTINUE
SchoolType
PRESCHOOL
PRIMARY
SECONDARY_GENERAL
SECONDARY_TECHNICAL
HIGHER_ED
NON_FORMAL
SchoolStatus
PENDING
VALIDATED
REJECTED
SUSPENDED
ExamType
SELF_ASSESSMENT
FORMATIVE_QUIZ
FINAL_EXAM
PRACTICAL_WORK
OLYMPIAD
MOCK_EXAM
MINI_TEST
CASE_STUDY
ExamStatus
DRAFT
PENDING_VALIDATION
VALIDATED
PUBLISHED
ARCHIVED
EvaluationType
QCM
TRUE_FALSE
OPEN_QUESTION
CASE_STUDY
EXAM_SIMULATION
RUBRIC
AttemptStatus
IN_PROGRESS
SUBMITTED
GRADED
ABANDONED
MediaType
VIDEO
PODCAST
AUDIO
BOOK
TransactionStatus
PENDING
PROCESSING
COMPLETED
FAILED
REFUNDED
SubscriptionStatus
FREEMIUM
PREMIUM
INSTITUTION_PREMIUM
EDUCATOR_ACCESS
DifficultyLevel
BEGINNER
INTERMEDIATE
ADVANCED
EXPERT
MasteryLevel
UNKNOWN
AWARE
FAMILIAR
UNDERSTANDING
APPLYING
MASTERED
PERFECTLY_ABLE
BadgeCategory
PERFORMANCE
STREAK
SOCIAL
EXPLORATION
MASTERY
ChallengeType
CLASS_BATTLE
SOLO_CHALLENGE
SCHOOL_LEAGUE
SUBJECT_SPRINT
ConversationStatus
ACTIVE
ARCHIVED
CLOSED
AssistanceRequestStatus
OPEN
IN_PROGRESS
RESOLVED
CLOSED
PayoutStatus
PENDING
PROCESSING
COMPLETED
FAILED
UEType
FONDAMENTALE
COMPLEMENTAIRE
TRANSVERSALE
STAGE
MEMOIRE
SpecialtyLevel
LICENCE
MASTER
DOCTORAT
BTS_HND
DeliveryMode
PRESENTIEL
EN_LIGNE
HYBRIDE
LateCodeStatus
ACTIVE
USED
EXPIRED
REVOKED
ImportStatus
PENDING
PROCESSING
COMPLETED
FAILED
DemandStatus
HIGH
MEDIUM
LOW
TutelleType
MINISTERIELLE
UNIVERSITAIRE
PRIVEE
ApprovalStatus
PENDING
APPROVED
REJECTED

Juin 2026 — 74 modèles + patterns lib/